ESP 411 - PRACTICAL 1

1. INTRODUCTION

The outset of this practical requires the implementation of the Fast Fourier transform to provide information of the frequency content of sampled audio signal (10Hz – 20 kHz). The information must be communicated to a device that displays the spectral information of the sampled signal. Figure 1 provides the flow diagram of the audio spectrum analyser. Figure 1 shows the typical implementation of the audio spectrum analyser.

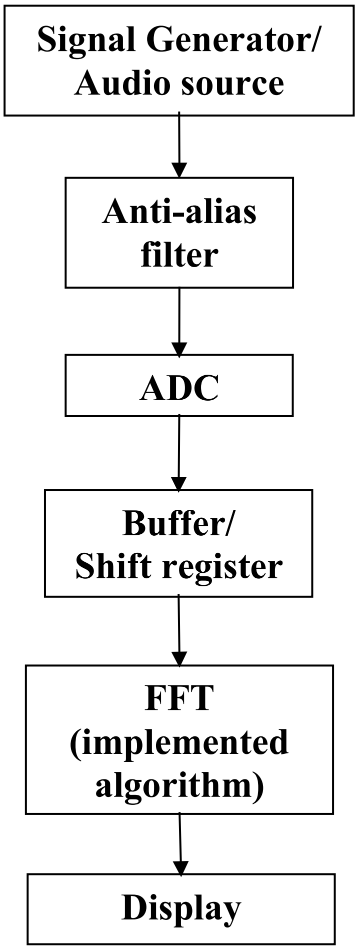

Figure 1. Typical implementation of the audio spectrum analyser

2. EQUIPMENT REQUIRED

The equipment required for successful completion of the practical would be the following:

- Audio input device (i.e. Signal generator, MP3 player or WAV file player). ***Each group must provide their own***.
- LF-347 (or similar) op-amp and other passive components (resistors, capacitors etc.)
- STM32F429 Discovery Board
- IDE and other essential software

Note: It is critical that students work through all documentation related to the STM32F429 DISCOVERY BOARD. A compulsory test on the material will be written on Tuesday 17 February 2026.

3. IMPLEMENTATION

3.1. Anti-alias filter

A 20 kHz anti-alias filter should be designed and implemented before sampling with the ADC converter (**triple interleaved is now required**).

3.2. FFT Implementation

The STM32F429 is to sample (ADC) and perform a 128 point (or greater) FFT on the sampled signal (0 – 1V). **The FFT must be implemented from first principle**. Your results can however be ***compared*** to the DSP Library's FFT function (CMSIS).

3.3. Output

The output device is to plot the frequency data result accurately. The magnitude information should be displayed on the DISCOVERY BOARD's graphical display (dB vs kHz).
Note: If you can get the DAC output to work and display the quantized signal, it will be beneficial towards Practical 2 (as it will be required).

4. EVALUATION PROCEDURE

4.1. Evaluation of Practical

The marks for the practical will be evaluated in the following manner:

Section	**Details**	**Mark**
Input Filter	Correct input of the sinusoidal signal by the ADC	10
ADC	Implementation of a triple interleaved ADC	10
DSP	Implementation of 128 point (or greater) FFT from first principle.	20
Display	Correct evaluation of FFT data and displaying of all relevant data.	10
Total		50

Final Mark (100) = Practical (50) + Documentation (50)

4.2. Documentation Deliverables

A technical report should be handed in on 5 March 2026 before 23:59 on the AMS system, with emphasis on the following:

- Design and simulation of the scaling circuit and analog filters (anti-alias and reconstruction).
- Implementation of DSP sampling and FFT.
- Software flow diagrams
- Commented code for both the FFT implementation and Display software.